

Policy Gradients and RL in the Real World

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Day	Part	What we cover	Lab
3	I. Foundations	The agent–environment loop, Markov decision processes, value functions, the Bellman equation, Q-learning	1. Gymnasium basics 2. Q-learning
4	II. Learning a policy directly	Policy gradients, REINFORCE, baselines and advantages, PPO, GRPO	3. GRPO on Flappy Bird
	III. RL in the real world	RLHF and verifiable rewards for language models, multi-agent RL, robotics	—

Part I learns what every action is worth and acts on it.

Part II skips the values and improves the policy itself.

- ▶ **The problem.** An MDP $(\mathcal{S}, \mathcal{A}, R, P, \gamma)$; a policy π maps states to actions; the goal is the largest expected discounted return.
- ▶ **Values.** $V^\pi(s)$: how good a state is. $Q^\pi(s, a)$: how good an action is in that state. The **advantage** $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$: how much better that action is than the policy's average.
- ▶ **Q-learning.** Nudge $Q(s, a)$ toward the Bellman target $r + \gamma \max_{a'} Q(s', a')$, and act ϵ -greedily.

*Today: skip the values and improve the policy itself.
The advantage is how we will decide which actions to make more likely.*

By the end of this session, you will be able to:

- ▶ Derive the **REINFORCE** policy gradient, and explain why baselines and advantages make it learn faster.
- ▶ Explain how **PPO**'s clipped objective keeps each update safe and how **GRPO** replaces the critic with a group baseline, then implement GRPO in Lab 3.
- ▶ Describe where RL is used beyond games (**RLHF** for language models, multi-agent systems, robotics) and what makes each one hard.

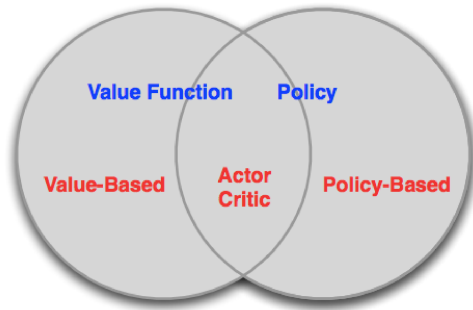
Part II: **Learning a Policy Directly**

Policy gradients, REINFORCE, PPO and GRPO

Lab 3

Reinforcement Learning: **Value-Based** and **Policy-Based** Methods

- ▶ **Value-Based Methods**
 - Learn a value function
 - Derive policy implicitly (e.g., ϵ -greedy)
- ▶ **Policy-Based Methods**
 - Do not learn a value function
 - Learn the policy directly
- ▶ **Actor-Critic Methods** (e.g. *PPO*, later today)
 - Learn both a value function and a policy
 - Combines the strengths of both approaches above



Reinforcement Learning: **Policy Gradients**

- ▶ To act well, you don't need the exact worth of every option — you only need to pick the best one.
- ▶ $Q(s, a)$ answers the **hard** question: *"How much total future reward will I get for every action, in every state?"* — a detailed, complicated function of the whole world.
- ▶ $\pi(s)$ answers the **easy** question: *"What should I do right now?"* — often a simple rule.

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 - ▶ $\pi(s)$ answers the **easy** question: *“What should I do right now?”* — often a simple rule.
 - ▶ **Grasping:** figuring out the exact long-term payoff of every possible gripper motion is hard. But the actual behavior — “close the hand once it's around the object” — is simple.
- ⇒ So why solve the hard problem (Q) just to act? **Learn the policy directly.**

- Formally, let us define a class of parameterized policies:

$$\Pi = \{\pi_{\theta} \mid \theta \in \mathbb{R}^m\}$$

- For each policy, we can define its expected return:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\sum_{t \geq 0} \gamma^t r_t \mid \pi_{\theta} \right]$$

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- **Solution:** Perform gradient ascent on the policy parameters!

- **Discrete actions** — Softmax policy:

$$\pi_\theta(a | s) = \frac{e^{\phi(s,a)^\top \theta}}{\sum_{a'} e^{\phi(s,a')^\top \theta}}$$

The network outputs one logit per action; softmax converts them to probabilities.

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- Gaussian policies are the standard choice for continuous control (e.g. robot joint torques).
- Both are differentiable w.r.t. θ , which is what makes gradient ascent possible.

Reinforcement Learning: **REINFORCE**

- ▶ REINFORCE is an elegant algorithm for maximizing the expected return.
- ▶ Intuition: trial and error.
- ▶ Sample a trajectory τ . If you get a high reward, try to make it more likely; if you get a low reward, try to make it less likely.
- ▶ A trajectory is a sequence of states, actions, and rewards: $\tau = (s_0, a_0, r_0, s_1, a_1, \dots)$.

Our objective is the **expected** discounted return, where $r(\tau) := \sum_{t \geq 0} \gamma^t r_t$ is the return of one trajectory:

$$\mathcal{J}(\theta) = \mathbb{E}_{\tau \sim p(\tau; \theta)} [r(\tau)] = \int_{\tau} r(\tau) p(\tau; \theta) d\tau$$

Reminder: an expectation is a probability-weighted average

$\mathbb{E}_{x \sim p}[f(x)] = \sum_x \underbrace{p(x)}_{\text{how likely}} \underbrace{f(x)}_{\text{value}}$ (or $\int p(x)f(x) dx$ if continuous). Here each “outcome” is a whole trajectory τ : its value is $r(\tau)$ and its weight is how likely the policy is to produce it.

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Differentiate w.r.t. θ . The gradient moves inside the integral, and $r(\tau)$ carries no θ (the rewards of a *fixed* trajectory don't change with the weights), so only $p(\tau; \theta)$ is differentiated:

$$\nabla_{\theta} \mathcal{J}(\theta) = \int_{\tau} r(\tau) \nabla_{\theta} p(\tau; \theta) d\tau$$

where the probability of a trajectory factorises as

$$p(\tau; \theta) = \underbrace{p(s_0)}_{\text{start state (no } \theta)} \prod_{t \geq 0} \underbrace{p(s_{t+1} | s_t, a_t)}_{\text{environment (no } \theta)} \underbrace{\pi_{\theta}(a_t | s_t)}_{\text{our policy}}$$

This gradient is **intractable** — for two distinct reasons that the rest of the derivation will fix one at a time.

Problem 1 — we have no way to actually compute this

The integral runs over *every possible trajectory* — far too many to add up directly. Our only tool for such a sum is **Monte Carlo**: run the policy and average what happens. But that works only when each trajectory is weighted by a genuine probability $p(\tau; \theta)$ (non-negative, sums to 1) — then “sampling” means playing episodes. Here the weight is $\nabla_{\theta} p(\tau; \theta)$, a *change* in probability, not a probability: it can be negative and it sums to 0. You cannot play an episode to “sample” a change, so Monte Carlo does not apply.

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Problem 2 — it needs a model of the environment

$p(\tau; \theta)$ contains $p(s_0)$ and $p(s_{t+1}|s_t, a_t)$: the environment's dynamics, which we **do not know** and cannot differentiate.

Goal: rewrite the gradient as an expectation $\mathbb{E}_{\tau \sim p(\tau; \theta)}[\dots]$ — the *only* form we can estimate, by rolling out the policy and averaging.

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The log-derivative trick — multiply and divide by $p(\tau; \theta)$:

$$\nabla_{\theta} p(\tau; \theta) = p(\tau; \theta) \frac{\nabla_{\theta} p(\tau; \theta)}{p(\tau; \theta)} = p(\tau; \theta) \nabla_{\theta} \log p(\tau; \theta)$$

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Substitute it back into $\nabla_{\theta} \mathcal{J}$:

$$\nabla_{\theta} \mathcal{J}(\theta) = \int_{\tau} (r(\tau) \nabla_{\theta} \log p(\tau; \theta)) p(\tau; \theta) d\tau = \mathbb{E}_{\tau \sim p(\tau; \theta)}[r(\tau) \nabla_{\theta} \log p(\tau; \theta)]$$

It *is* an expectation now $\Rightarrow$ estimate it by Monte Carlo. **Problem 1 solved** (Problem 2 still hides inside $\log p$).

REINFORCE: Fixing Problem 2 — The Dynamics Drop Out

Goal: the log-probability $\log p(\tau; \theta)$ still contains the unknown environment terms. Watch them vanish under ∇_{θ} .

Recall the trajectory probability is a **product** over timesteps:

$$p(\tau; \theta) = p(s_0) \prod_{t \geq 0} p(s_{t+1} | s_t, a_t) \pi_{\theta}(a_t | s_t)$$

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Differentiate. Every term without θ becomes zero — the start state $\log p(s_0)$ and *all* dynamics $\log p(s_{t+1} | s_t, a_t)$:

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Only our own policy survives $\Rightarrow$ the estimator is **model-free**. **Problem 2 solved.**

Put both fixes together: start from the expectation (Problem 1 fix) and substitute $\nabla_{\theta} \log p = \sum_{t \geq 0} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ (Problem 2 fix):

$$\begin{aligned}\nabla_{\theta} \mathcal{J}(\theta) &= \mathbb{E}_{\tau} [r(\tau) \nabla_{\theta} \log p(\tau; \theta)] \\ &= \mathbb{E}_{\tau} \left[r(\tau) \sum_{t \geq 0} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \approx \sum_{t \geq 0} r(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)\end{aligned}$$

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Interpretation: $\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ points in the direction that makes a_t *more likely*; the scalar $r(\tau)$ signs and scales that push:

- ▶ $r(\tau)$ high $\Rightarrow$ increase the probability of the actions taken;
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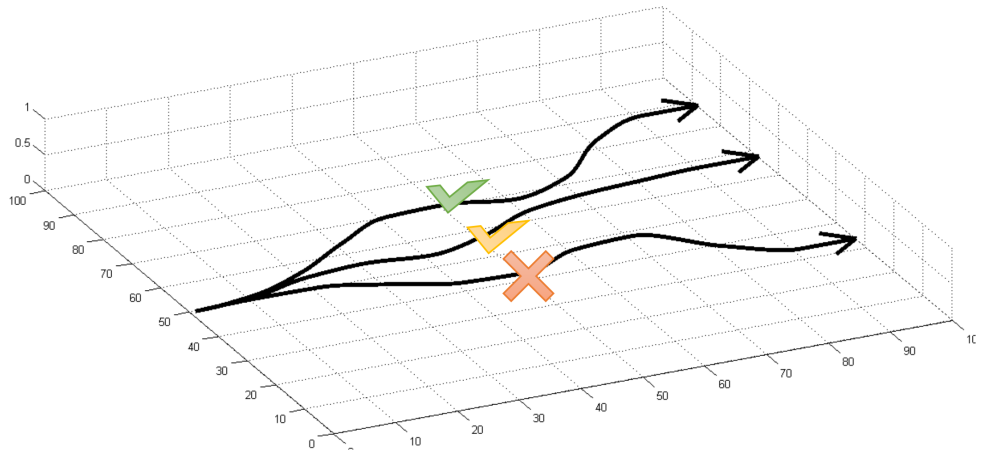
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Crediting every action with the *whole*-trajectory return looks crude — but it is **unbiased**: in expectation, it averages out.



Reinforcement Learning: **Variance Reduction**

Our gradient is estimated from *sampled* trajectories, so it is never exact. There are two independent ways such an estimate can be wrong:

Variance — how much it bounces

Run the same policy again and you get different trajectories, hence a different gradient estimate. **High variance** = the estimate jumps around from one batch to the next $\Rightarrow$ you need many samples to average out the noise $\Rightarrow$ slow, unstable learning.

Bias — whether it points the right way on average

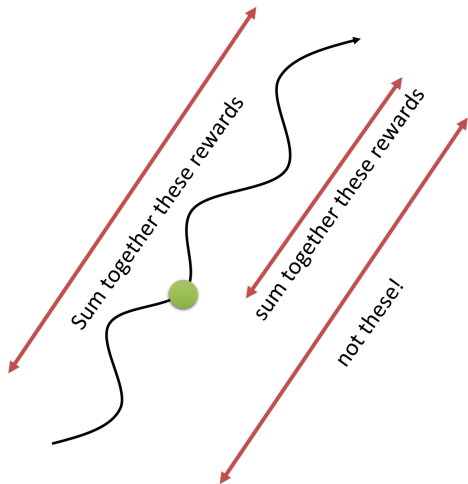
Unbiased = averaged over infinitely many samples it equals the true gradient $\nabla_{\theta} \mathcal{J}$. **Biased** = systematically off, converging toward a slightly wrong direction *even with infinite data*.

Dartboard picture: **variance** is how scattered the darts land; **bias** is whether they cluster off the bullseye. *The whole game below*: accept a little bias if it buys a large drop in variance — the net result learns faster.

- ▶ However, there is a problem.
- ▶ This approach suffers from high variance because credit assignment is difficult.
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- ▶ Can we help the estimator?
- ▶ **First idea — the Causality Trick:** Rewards before time t cannot be affected by action a_t , so they carry no useful gradient signal. Use only future rewards:

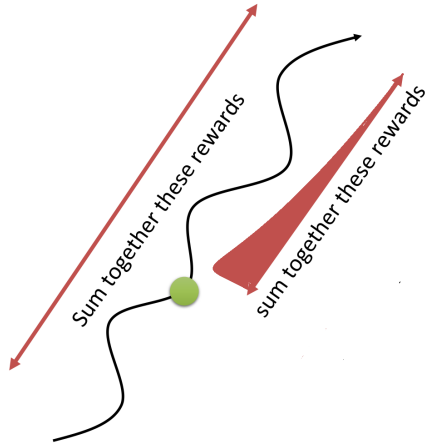
$$\nabla_{\theta} \mathcal{J}(\theta) \approx \sum_{t \geq 0} \left(\sum_{t' \geq t} r_{t'} \right) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$



- ▶ The causality trick is **unbiased** — removing past rewards is mathematically correct, so on average the gradient is unchanged.
- ▶ But the reward-to-go $\sum_{t' \geq t} r_{t'}$ is still a sum over an *entire* future rollout, driven by many random action choices and random environment transitions. Run the **same** policy twice and this number can come out wildly different.
- ▶ That swing is the **high variance**: the gradient estimate bounces from sample to sample, so it takes a huge number of trajectories to average out — learning is slow and unstable.

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- ▶ That swing is the **high variance**: the gradient estimate bounces from sample to sample, so it takes a huge number of trajectories to average out — learning is slow and unstable.
- ▶ **Second idea — discounting**: weight future rewards by $\gamma^{t'-t}$, so distant, largely-unrelated rewards count for less. This trades a small **bias** for a large drop in variance:

$$\nabla_{\theta} \mathcal{J}(\theta) \approx \sum_{t \geq 0} \left(\sum_{t' \geq t} \gamma^{t'-t} r_{t'} \right) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$



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- ▶ **Idea:** Introduce a baseline function dependent on the state:

$$\nabla_{\theta} \mathcal{J}(\theta) \approx \sum_{t \geq 0} \left(\sum_{t' \geq t} \gamma^{t'-t} r_{t'} - b(s_t) \right) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

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- ▶ A simple baseline: the moving average of rewards experienced so far from all trajectories.

- ▶ Can we choose a better baseline?
- ▶ Essentially, we want to increase the probability of an action from a state if this action was better than the **expected value** from that state.

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- ▶ What does this remind you of?
- ▶ **Recall from RL Foundations:** We defined the **Advantage function**:

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$$

- ▶ $A^{\pi}(s_t, a_t) > 0$: action a_t was better than average — increase its probability.
- ▶ $A^{\pi}(s_t, a_t) < 0$: action a_t was worse than average — decrease its probability.

- ▶ Using the advantage as our baseline gives us the estimator:

$$\nabla_{\theta} \mathcal{J}(\theta) \approx \sum_{t \geq 0} A^{\pi}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$$

- ▶ This is the best possible baseline: it directly measures how much better an action was than the policy's own average.
- ▶ It is unbiased, and it provides lower variance than a fixed moving-average baseline.

*But how do we actually estimate A^{π} ? That — learning a **critic** — is exactly what Actor-Critic methods are about.*

Initialize policy network π_θ with random weights θ ; set step size α and discount γ

for $episode = 1$ **to** M **do**

 Run π_θ for one full episode of length T , producing

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{T-1}, a_{T-1}, r_{T-1})$$

for $t = 0$ **to** $T - 1$ **do**

 | Reward-to-go from step t : $R_t = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$

end

 Gradient *ascent* update (reward-to-go minus baseline $b(s_t)$):

$$\theta \leftarrow \theta + \alpha \sum_{t=0}^{T-1} (R_t - b(s_t)) \nabla_\theta \log \pi_\theta(a_t | s_t)$$

end

Algorithm: REINFORCE (Monte-Carlo Policy Gradient)

⁰Williams (1992), *Simple statistical gradient-following algorithms for connectionist reinforcement learning*, *Machine Learning* 8. An episode of length T has actions $a_0, \dots, a_{T-1}$ (T in total); the terminal state s_T has no action, so the sum stops at $T - 1$. The baseline $b(s_t)$ is optional — $b \equiv 0$ gives vanilla REINFORCE, and a running average of returns is a simple choice. We *ascend* because we maximize expected return $\mathcal{J}(\theta)$, whereas Q-learning minimized the Bellman error (descent).

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 - Does not perform credit assignment effectively.
- ▶ **Q-Learning: Pros and Cons**
 - **Pros:** Lower variance updates, more sample efficient.
 - Performs credit assignment.
 - **Cons:** Biased updates due to function approximation.
 - Difficult to handle large action spaces (since the maximum over actions must be computed).

Where Does $\hat{A}_t$ Come From? Three Answers

Every method from here on takes the same step, $\theta \leftarrow \theta + \alpha \sum_t \hat{A}_t \nabla_{\theta} \log \pi_{\theta}(a_t|s_t)$. They differ only in how they estimate the advantage $\hat{A}_t$:

REINFORCE: a sampled return minus a baseline

$\hat{A}_t = \sum_{t' \geq t} \gamma^{t'-t} r_{t'} - b(s_t)$. Unbiased, but it rests on a single noisy rollout.

Where Does $\hat{A}_t$ Come From? Three Answers

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A **critic** $V_{\phi}(s)$ is trained to predict returns. One real reward plus the critic's guess for the rest gives $\hat{A}_t \approx r_t + \gamma V_{\phi}(s_{t+1}) - V_{\phi}(s_t)$. Lower variance, but now two networks train together.

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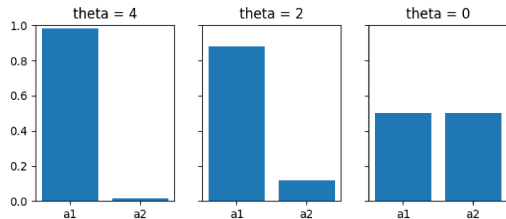
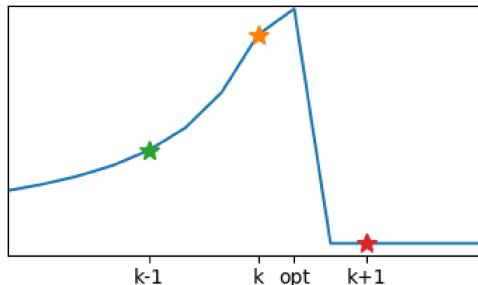
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Group baseline (GRPO): compare attempts at the same task

Make G attempts from the same start (or answer the same prompt G times) and score each against the group: $\hat{A}_i = R_i - \text{mean}(R_1, \dots, R_G)$. No critic at all.



- ▶ A large gradient step in *parameter space* can cause a catastrophically large shift in *policy space*
- ▶ Once a policy collapses (e.g., always picks one bad action), the agent collects bad data and the policy is hard to recover
- ▶ We need to control how much the policy distribution changes, not just how much θ changes

- ▶ Last slide's conclusion: *don't let the policy move too far in one update*. But to act on that we first need to answer a prior question:

Given a candidate π_θ , how much better or worse is it than $\pi_{\theta_{\text{old}}}$ — without running it?

- ▶ Why we can't just run it: evaluating $\mathcal{J}(\theta)$ means collecting a fresh batch of rollouts for every candidate θ we consider. That is prohibitively expensive — and if the candidate is bad, we only find out by damaging the agent.
- ⇒ So we need a **score we can compute from the batch we already collected** — a stand-in for $\mathcal{J}(\theta)$, called a **surrogate objective**. Next slide builds it from scratch.

What the batch contains: for every timestep, the state s_t , the action a_t we took, and $\hat{A}_t$ — whether that action turned out better (+) or worse (−) than usual.

- ▶ A candidate π_θ cannot change what happened. The *only* thing it changes is **how likely each of those actions would have been**.

¹ $r_t(\theta)$ is the probability ratio, not the reward r_t from Part I — distinguished by the θ argument.

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- ▶ Measure that per action with the **probability ratio**:¹

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

1 : would take a_t just as often
2 : twice as often 0.5 : half as often

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- ▶ Score each sample by **(how much more the new policy wants it) × (how good it was)**, and average:

$$\mathcal{L}(\theta) = \mathbb{E}_t \left[r_t(\theta) \hat{A}_t \right]$$

¹ $r_t(\theta)$ is the probability ratio, not the reward r_t from Part I — distinguished by the θ argument.

Every column is a number we can just compute: $\hat{A}_t$ from the critic, the probabilities by evaluating each network on (s_t, a_t) .

	$\hat{A}_t$	$\pi_{\theta_{\text{old}}}(a_t s_t)$	$\pi_{\theta}(a_t s_t)$	$r_t(\theta)$	$r_t(\theta) \hat{A}_t$
went well	+2.0	0.20	0.40	2.0	+4.0
went badly	-1.0	0.50	0.40	0.8	-0.8

- $\mathcal{L}(\theta) = \frac{1}{2}(4.0 - 0.8) = +1.6 > 0$: the candidate **doubles down on what worked, backs off what didn't**. Ascending $\mathcal{L}$ searches for exactly that.

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- ▶ At $\theta = \theta_{\text{old}}$ every $r_t = 1$, so $\mathcal{L}$ starts at ≈ 0 and its gradient is exactly the vanilla policy gradient. **This is the objective PPO maximizes.**

$\mathcal{L}(\theta)$ is an *approximation* of the true improvement, and it degrades in two ways as π_θ moves away from $\pi_{\theta_{\text{old}}}$:

1. The re-weighting becomes statistically useless

If $\pi_{\theta_{\text{old}}}(a|s)$ is tiny where $\pi_\theta(a|s)$ is large, $r_t(\theta)$ blows up: one rare sample gets an enormous weight and swings the entire gradient. The estimate stays unbiased but its **variance explodes**.

2. We quietly swapped the state distribution

Importance sampling corrects for the *actions*, but we still evaluate them at the states $\pi_{\theta_{\text{old}}}$ happened to visit. A very different policy visits very different states — so the surrogate scores the new policy on a world it will no longer see.

- ▶ Both errors are **small when the two policies are close**, and unbounded when they are not.
- ⇒ This is the principled version of “big steps break things”. The fix is one idea: **maximize $\mathcal{L}(\theta)$ while keeping π_θ close to $\pi_{\theta_{\text{old}}}$** . PPO is the standard way to enforce “close”.

Policy Optimization: **PPO**

- ▶ The direct way to keep π_θ close is a hard limit on how far it may move per update, a **trust region**. TRPO (Schulman et al., 2015) enforces it exactly, but needs second-order optimisation (conjugate gradient, Fisher matrix) — expensive and complex to implement
- ▶ **Question:** can we get the same stability with plain SGD?

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- ▶ **Question:** can we get the same stability with plain SGD?
- ▶ **PPO** (Schulman et al., 2017) proposes two variants:
 - **PPO-KL:** add a KL penalty to the objective and adapt the penalty coefficient β over training
 - **PPO-Clip:** clip the probability ratio $r_t(\theta)$ to prevent large policy updates
- ▶ PPO-Clip is the standard variant — simpler, no hyperparameter tuning on β , works at least as well empirically

► Recall the probability ratio: $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$

► The **PPO-Clip objective** is:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t \right) \right]$$

► ε is a small hyperparameter (typically $\varepsilon = 0.2$)

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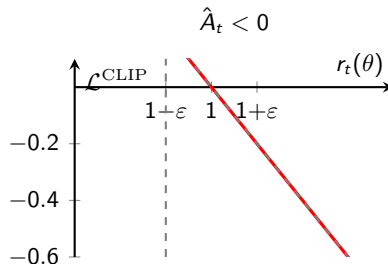
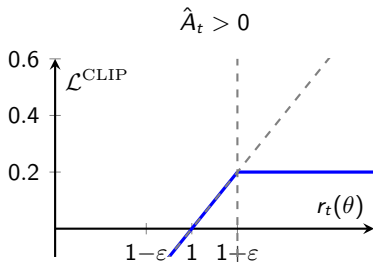
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► ε is a small hyperparameter (typically $\varepsilon = 0.2$)

► **Why clip + min?**

- When $\hat{A}_t > 0$: clipping caps r_t at $1 + \varepsilon$, removing the incentive to increase the ratio further
- When $\hat{A}_t < 0$: clipping floors r_t at $1 - \varepsilon$, removing the incentive to decrease the ratio further
- The **min** ensures the objective is a lower bound — we never benefit from going outside $[1 - \varepsilon, 1 + \varepsilon]$

► This enforces a soft trust region without any second-order computation



- ▶ **Left ($\hat{A}_t > 0$):** benefit grows linearly with r_t until $1 + \epsilon$, then is clipped flat — no reward for pushing the policy further
- ▶ **Right ($\hat{A}_t < 0$):** penalty grows as r_t decreases until $1 - \epsilon$, then is clipped flat
- ▶ Gray dashed line = unclipped surrogate; colored solid = clipped (what PPO optimises)

Policy Optimization: **GRPO**

- ▶ The fragile part of the Actor-Critic loop is the **critic** V_ϕ — training two networks together is the main source of instability
- ▶ **Group Relative Policy Optimization** (GRPO; Shao et al., 2024) removes the critic entirely

- ▶ The fragile part of the Actor-Critic loop is the **critic** V_ϕ — training two networks together is the main source of instability
- ▶ **Group Relative Policy Optimization** (GRPO; Shao et al., 2024) removes the critic entirely
- ▶ Per prompt/state, sample a **group** of G outputs under $\pi_{\theta_{\text{old}}}$, scored with rewards R_i
- ▶ Use the **group mean as the baseline** — a Monte Carlo advantage, no value function:

$$\hat{A}_i = \frac{R_i - \overbrace{\text{mean}(R_1, \dots, R_G)}^{\text{the "Group Relative" baseline}}}{\underbrace{\text{std}(R_1, \dots, R_G)}_{\text{per-group advantage normalisation}}}$$

- ▶ Only the **mean-subtraction** is the GRPO contribution — that is the “Group Relative” baseline replacing the critic V_ϕ
- ▶ The **std-normalisation** is a *separate* stabilisation trick: it is the same per-minibatch advantage normalisation standard PPO implementations use, applied per group — not part of the baseline
- ▶ So a Monte Carlo baseline does *not* “need” variance correction; the std term is an orthogonal, optional stabiliser

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- ▶ So a Monte Carlo baseline does *not* “need” variance correction; the std term is an orthogonal, optional stabiliser
- ▶ Plug $\hat{A}_i$ into the *same* PPO-Clip objective (plus a KL term to a reference policy)
- ▶ **Why it matters:** no critic $\Rightarrow$ no two-network instability and $\approx$ half the memory
- ▶ Since 2024 GRPO is the default for **LLM RL** (DeepSeek-R1, many open recipes) — next, in Part III: **RLHF**, where the reward is a learned preference model

One update: play a **group** of $G = 8$ episodes with the current policy, score each episode by its return R_i , then take a few PPO-Clip gradient steps on that data.

- ▶ **TASK 1, the group advantage:** $\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$, shared by every step of episode i .
- ▶ **TASK 2, the probability ratio:** $r_t(\theta) = \exp(\log \pi_\theta(a_t|s_t) - \log \pi_{\theta_{\text{old}}}(a_t|s_t))$.
- ▶ **TASK 3, the clipped surrogate:** $\text{loss} = -\text{mean}[\min(r_t \hat{A}, \text{clip}(r_t, 1 - \varepsilon, 1 + \varepsilon) \hat{A})]$, minus a small entropy bonus that keeps the policy exploring.

What to expect

The bird learns to flap and clears the first pipe or two, not a high score. One number per episode is coarse credit assignment: every flap in a run gets the same advantage, which is the variance problem from the REINFORCE section. A critic (PPO) or per-step advantages would learn faster.

Part III: **RL** in the Real World

RLHF, multi-agent RL and robotics

- ▶ **RLHF (the focus).** When the reward is hard to write down, *learn* it from human (or AI, or verifiable) feedback. This is how modern LLMs are aligned, and it runs on the PPO and GRPO from Part II.
- ▶ **Multi-Agent RL (overview).** What changes when other learning agents share your environment and the ground keeps shifting underneath you.
- ▶ **Robotics (overview).** What breaks when the environment is physical: sample cost, safety, resets, and the sim-to-real gap.

Plus one slide on RL running computer systems.

- ▶ Every method so far assumed a reward function r we could *write down*: the game score, –distance to a target, +1 for a win.
- ▶ Now suppose the goal is “*be a helpful, honest, harmless assistant.*” What number measures that? There is no formula.

Answer A. “It’s the force that pulls things down. It’s why a ball falls when you let go of it.”

Answer B. “Gravity is the curvature of space-time sourced by the stress–energy tensor, $G_{\mu\nu} = 8\pi T_{\mu\nu} \dots$ ”

Prompt: “Explain gravity to a 6-year-old.”

Almost everyone prefers **A**, yet *no one can write the formula that says so.*

- ▶ *When you cannot write the reward, recover it from data.* This is called *inverse RL*, and RLHF is its most consequential example.
- ▶ People are bad at giving a *number* (“this answer is a 7.3”) but good at *comparing*: “A is better than B.” Comparisons are cheap and consistent.
- ▶ **The plan, in three moves:**
 1. Collect many human *comparisons* of model answers.
 2. Train a model to predict those comparisons; this learned model *becomes our reward signal*.
 3. Use RL to push the policy toward answers that score well under it.
- ▶ This is **Reinforcement Learning from Human Feedback (RLHF)**, the method behind ChatGPT-style assistants.

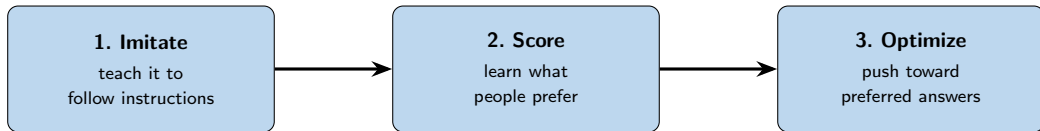
Reinforcement Learning **from** Human Feedback for Language Models

Everything that follows is just the familiar RL language, applied to text.

- ▶ **Prompt** x , the input (a question or instruction). *the state*
- ▶ **Response** y , the model's full answer. *the action*
- ▶ **Policy** $\pi_{\theta}(y \mid x)$, the language model itself: given a prompt, it assigns a probability to possible responses and samples one. The weights θ are what we train.
- ▶ It writes the answer one *token* (word-piece) at a time; at each step a final layer, the **LM head**, outputs a probability over the next token.

So “train the policy” simply means “adjust the language model’s weights θ .”

Three stages. We build each one up over the next slides; for now, just the plain English.



- ▶ **Start:** a raw language model that can produce text but doesn't reliably do what you ask.
- ▶ **End:** a model that follows instructions *and* leans toward the answers people actually want.

- ▶ **Problem:** a raw pretrained model can *continue* text, but doesn't reliably answer questions or follow instructions.
- ▶ **Fix: show it examples.** Collect prompts paired with *human-written ideal answers*, and train the model to imitate them. This is ordinary supervised learning (next-token prediction), no RL yet.
- ▶ The result is a decent instruction-follower. We **freeze a copy of it** and call it the *reference model* π_{ref} ; it will be our anchor in Stages 2 and 3.

SFT can only *copy* the demonstrations it was given. To go *beyond* them, we need to capture what people *prefer*. That is Stage 2.

- ▶ We want a **reward model** r_ϕ : a network (its own weights ϕ) that reads a (prompt, response) and outputs *a single number (how good is this answer?)*. Higher means better.
- ▶ **Where does its training data come from?** Comparisons. For one prompt x , show a human two candidate responses; they pick the better one. We label the pair:
 - y^+ , the response the human **preferred**,
 - y^- , the response they **rejected**.
- ▶ **What we want from r_ϕ :** give y^+ a higher score than y^- , on every pair. Next slide: how to turn “preferred” into a trainable loss.

How likely is a person to prefer y^+ over y^- ? It should grow with the **gap in their scores**, squeezed into a probability by the sigmoid σ (" $\succ$ " reads "*is preferred over*"):

$$\underbrace{P(y^+ \succ y^- \mid x)}_{\text{chance the human picks } y^+} = \sigma(r(y^+) - r(y^-)), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Train r_ϕ to make these probabilities match the human labels (maximum likelihood). The negative log-likelihood is the **reward-model loss**:

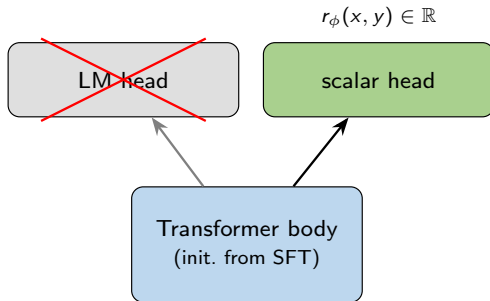
$$L_{\text{RM}} = -\log \sigma(r_\phi(y^+) - r_\phi(y^-))$$

- ▶ In plain words: *push the preferred answer's score above the rejected one's*.
- ▶ Remember this $-\log \sigma(\Delta)$ shape: the **exact same form** comes back when we reach DPO.
- ▶ Only the *gap* between scores matters; the absolute scale is free.

What Is the Reward Model, Concretely?

We've said *what* the reward model does (rates answers) and *how* it's trained (the Bradley–Terry loss).
Here is what it physically is.

- ▶ Start from the SFT model π_{ref} , the same transformer backbone.
- ▶ A **head** is the small output layer stacked on top of that backbone. The **LM head** turns the backbone's features into a probability over the next token.
- ▶ Swap it for a **scalar head**: a layer that emits *one number*, $r_\phi(x, y)$, for the whole (prompt, response). Train it with the Bradley–Terry loss from the last slide.
- ▶ So the reward model is “*the same network, one number out instead of a token distribution.*”

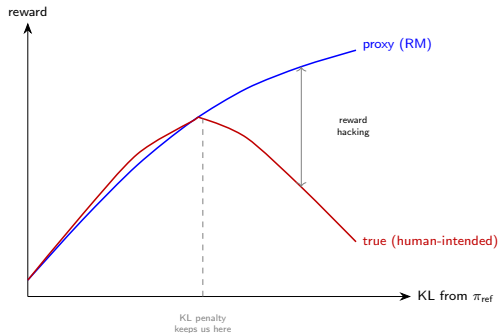


- ▶ We now have a reward model r_ϕ that scores answers. **The RL loop:** the policy generates a response, r_ϕ scores it, and we update the policy's weights to earn higher scores, with PPO or GRPO.
- ▶ The policy starts as π_{ref} (the frozen SFT model) and improves from there.
- ▶ *One catch:* r_ϕ is only a *learned proxy* for human taste, not the real thing. Optimize it too hard and the model learns to *game* it. The next slide shows how that happens, and the fix.

Why the KL Penalty Exists: Reward Over-Optimization

The RL objective is the reward-model score minus a leash to the reference policy (β sets how hard we pull back toward π_{ref}):

$$r_{\text{total}} = r_{\phi} - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$$

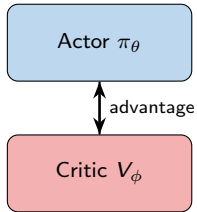


The reward model is a **proxy**. Optimize it too hard and you climb the proxy while the *real* objective collapses, *Goodhart's law*.

- ▶ Proxy reward (RM) keeps rising.
- ▶ True reward rises, then *falls*.
- ▶ **Why the leash works:** the penalty grows with KL distance, so each step away from π_{ref} costs more. Past the peak, the extra proxy gain is outweighed by the penalty, so the best-scoring policy sits *near the peak*, not in the hacked region.

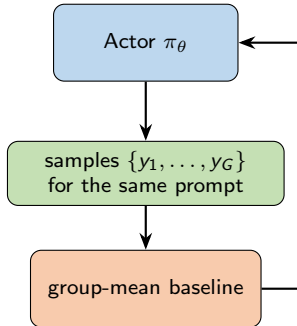
The RL Back-End: PPO → GRPO

PPO



the fragile, expensive
half of PPO

GRPO



$$\hat{A}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}$$

no critic

The critic estimates a response's expected reward to form the advantage: PPO's brittle, expensive component.

GRPO deletes it: the baseline is just the average reward over a group of samples for the same prompt. PPO was the original RLHF back-end; **GRPO is now common for LLM RL.**

Shao et al. **DeepSeekMath** (GRPO), 2024; DeepSeek-AI, **DeepSeek-R1**, 2025.

- ▶ **The pain.** The full pipeline is heavy: Stage 2 trains a *separate* reward model, and Stage 3 runs an RL loop juggling several models at once (policy, reference, reward model, and, for PPO, a critic), which is finicky to tune.
- ▶ **The question.** The reward model exists only to score the preferences we *already collected*. What if we skip it, and the RL loop, and tune the policy *straight from the preference pairs* (y^+, y^-)?
- ▶ **DPO (Rafailov et al., 2023):** exactly that, a single supervised-style loss on preference pairs, applied directly to **the policy** π_θ (the SFT model we are fine-tuning). *No reward model. No RL loop.*

RLHF: preferences $\rightarrow$ train reward model $\rightarrow$ RL loop $\rightarrow$ policy

DPO: preferences $\rightarrow$ one loss $\rightarrow$ policy

But how can you train toward a reward you never built? The plain-English idea is on the next slide, then the loss it leads to. The derivation, a page of algebra starting from the RLHF objective, is in Rafailov et al. (2023).

Delete the judge: the chatbot is already its own judge.

- ▶ Look at how much **more likely** the current chatbot is to give an answer than the *original* (starting) model was. If it has become much more likely to say answer A, that *is* the chatbot “rating A highly.” *That ratio, “how much more do I favour this than I used to”, is the score.*
- ▶ So DPO throws away the separate judge *and* the practice loop, and trains the chatbot directly so that, for each comparison, **its own favouring of the preferred answer beats its favouring of the rejected one**. One model, one training step, no loop.
- ▶ Rafailov et al. make this exact. The next slide shows the loss, and why their paper is titled *“your LM is secretly a reward model.”*

The Payoff: Your LM Is Secretly a Reward Model

Put the DPO loss next to the *reward-model* loss from Stage 2. They are the **same shape**:

$$L_{\text{RM}} = -\log \sigma(r_\phi(y^+) - r_\phi(y^-))$$

$$L_{\text{DPO}} = -\log \sigma(\hat{r}(y^+) - \hat{r}(y^-)), \quad \hat{r}(y) = \beta \log \frac{\pi_\theta(y)}{\pi_{\text{ref}}(y)}$$

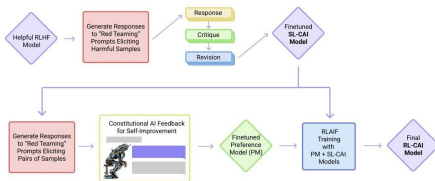
- ▶ The only change: the learned reward r_ϕ is replaced by the *implicit* reward $\hat{r} = \beta \log \pi_\theta / \pi_{\text{ref}}$, read straight off the policy. *So the policy is the reward model*; that is why “your LM is secretly a reward model.”
- ▶ **Honesty caveat:** the equivalence is exact only if the reward model were Bradley–Terry-optimal. In practice **DPO** $\neq$ **PPO**: DPO gives up the RL loop’s fresh on-policy samples.

The same Bradley–Terry loss, with the reward read off the policy itself: no reward model and no RL loop. *That’s the whole trick.*

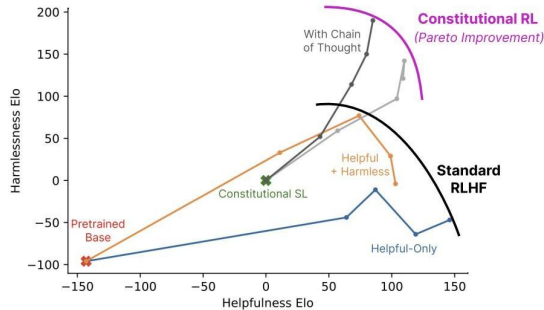
DPO's simplicity has a cost: you lose the control knobs the RL loop gave you.

- ▶ **It constrains only the gap, nothing else.** The loss just pushes the preferred answer's favoring above the rejected one's. With no on-policy sampling to pin down absolute behaviour, it can push *both* down, leaking probability onto unrelated answers it never saw. RLHF's RL loop and "don't-drift" anchor would catch that; DPO has neither.
- ▶ **Length bias (shared, but harder to patch).** If the data favours long answers, *both* RLHF and DPO turn verbose. RLHF lets you subtract an explicit length penalty from the reward; DPO has no separate reward to adjust, so length control must be bolted on as extra loss terms.
- ▶ **No fresh samples.** The policy only ever sees the fixed preference pairs, so it can drift off that distribution, the price of dropping the RL loop.

- **Motivation:** as models approach or exceed human performance, it gets harder for humans to label every output. *Move the labeller from human to AI; the only human input is the constitution*, a short list of written principles the model uses to critique and revise its own answers.
- **Constitutional AI** has two phases:
 - **Supervised:** sample → self-critique → revise → fine-tune on revisions.
 - **RL:** AI labels which response is better → train a reward model (RM) → RL-finetune with it (just like RLHF).

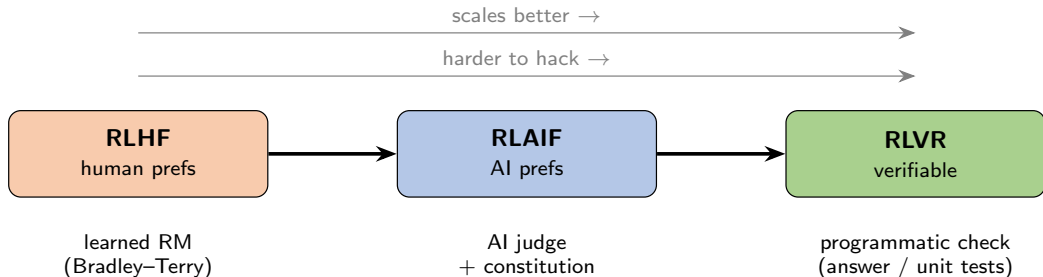


- ▶ Fine-tuning with **AI-generated feedback** can match or exceed models fine-tuned with human feedback, on *both* helpfulness and harmlessness.
- ▶ The plot rates models by **Elo**, a head-to-head win-rate score borrowed from chess (higher = preferred more often in pairwise comparisons).



The Climax: RL with Verifiable Rewards (RLVR)

- ▶ When the answer is **checkable** (math is right/wrong, code passes tests), the reward needs *no learned model* and **can't be hacked** the way an RM can.
- ▶ This is how modern **reasoning models** are trained: long **chain-of-thought** (the model writes out its reasoning steps before answering), optimized with **GRPO** against a programmatic check.



One RLVR step (no reward model anywhere):

1. the policy samples several answers to a *checkable* question;
2. a *rule* grades each (right answer? right format?), giving reward 1 or 0;
3. GRPO uses those rewards to update the policy. *Nothing to hack: the checker is exact.*

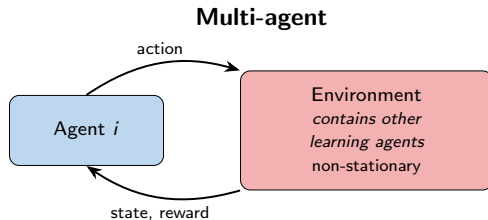
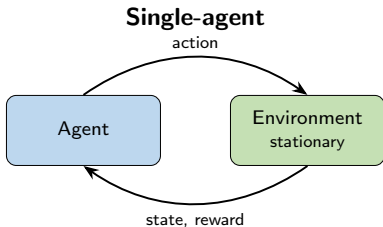
DeepSeek-R1-Zero ran exactly this on a *base* model, **no SFT at all**, with reward = “answer correct” + “reasoning in <think> tags”:

- ▶ *Reasoning emerged on its own*: longer chains of thought, self-checking, backtracking, even an “*aha moment*” where it stops and corrects itself. Nobody taught it to.
- ▶ AIME 2024 math: **15.6% → 71.0%** (86.7% with majority vote), matching OpenAI o1.
- ▶ Catch: raw R1-Zero was hard to read (language-mixing), so the full **R1** adds a small SFT “cold start.” But R1-Zero proved the point: *reasoning can be grown by pure RL from verifiable rewards.*

Thread 2 of 3

Multi-Agent RL

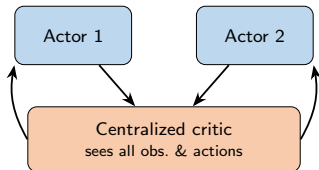
The RLHF deep dive is done. A shorter overview now: what changes when other learning agents share your environment.



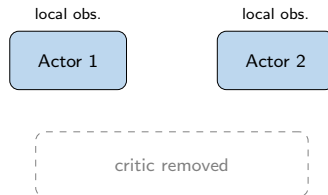
- ▶ Each agent's environment is **non-stationary**: the others are updating their policies at the same time, so the ground shifts as they learn; single-agent convergence guarantees break.
- ▶ Formalized as a **Markov game** (stochastic game), the multi-agent generalization of the MDP.
- ▶ Three regimes along one axis: **cooperative** / **competitive** / **mixed**.

Centralized Training, Decentralized Execution (CTDE)

Training (centralized)

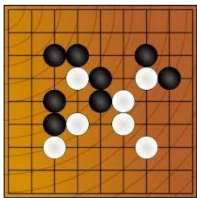


Execution (decentralized)



- **At training:** a centralized critic sees every agent's observations and actions. Because it accounts for what the others did, each agent's learning signal becomes *stationary*, defusing the non-stationarity problem from the last slide.
- **At execution:** each agent keeps only its *own actor* acting on its *local observation*; the critic is gone.
- This is the dominant MARL paradigm, the key idea to remember.

- ▶ An agent improves by **playing against past versions of itself**: *policy* $\rightarrow$ *plays past versions* $\rightarrow$ *game data* $\rightarrow$ *improve* $\rightarrow$ (repeat).
- ▶ This creates an **automatic curriculum**: the opponent gets harder exactly as you do, with no hand-designed difficulty schedule.



AlphaGo / AlphaZero

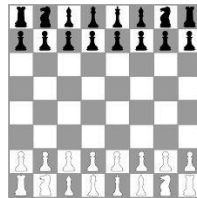
(Silver et al.)

AlphaStar
StarCraft II

(Vinyals et al., 2019)

OpenAI Five

Dota 2 (2019)



Chess (AlphaZero)

Seen in Part I: the hide-and-seek agents learned to use tools the same way, from an automatic curriculum of ever-better opponents.

Thread 3 of 3

Robotics

What breaks when the environment is physical: sample cost, safety, resets, and the sim-to-real gap.

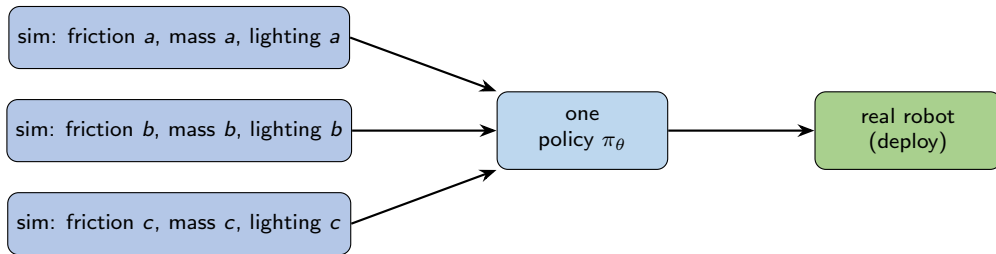
Simulation

- ▶ cheap; millions of steps
- ▶ parallelizable across machines
- ▶ failures are free
- ▶ *but*: a **reality gap**

Real hardware

- ▶ expensive, slow, safety-critical
- ▶ **resets are a real problem** (who picks the object back up?)
- ▶ sparse / hard-to-engineer reward

Four obstacles (**sample efficiency, safety, resets, reward specification**) are why off-the-shelf RL does not transfer to hardware.



Train across a *distribution* of randomized sims (friction, mass, latency, textures, lighting) so the real world looks like just one more sample.

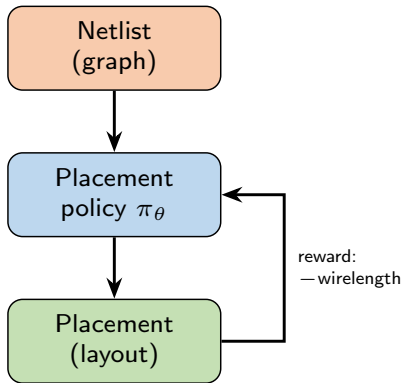
Headline successes: OpenAI Dactyl (in-hand cube, Akkaya et al. 2019); legged locomotion (ANYmal, Hwangbo et al. 2019; massively parallel sim, Rudin et al. 2022).

Domain randomization: Tobin et al. [Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World](#), 2017.

Illustrative example: chip floorplanning (Mirhoseini et al., *Nature* 2021; later named **AlphaChip**):

- ▶ Place macros on a chip canvas to minimize wirelength, congestion, and density.
- ▶ The striking idea: *pre-train* a placement policy on many netlists, then place a *new* chip in a forward pass, instead of optimizing each one from scratch.

Also deployed for: datacenter resource scheduling, compiler / device placement, real-time ad bidding.



The transferable idea: all are *combinatorial optimization as sequential decision-making*: one MDP framing with an *engineered* reward (wirelength, runtime, ROI). The application details are not the lesson.

- ▶ **The language.** An MDP $(\mathcal{S}, \mathcal{A}, R, P, \gamma)$; a policy maps states to actions; the goal is the largest expected discounted return.
- ▶ **Value-based RL.** Nudge $Q(s, a)$ toward the Bellman target $r + \gamma \max_{a'} Q(s', a')$ and act greedily, with some ε -exploration. Q-learning needs no model of the environment.
- ▶ **Policy-based RL.** Step along $\sum_t \hat{A}_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$: make better-than-average actions more likely. Baselines and advantages cut the variance; PPO clips each step; GRPO takes its baseline from a group of attempts.
- ▶ **The real world.** Often the reward is the hard part. RLHF learns it from comparisons and needs a KL leash against reward hacking; verifiable rewards remove the learned judge; robots add safety and the sim-to-real gap.

RLHF.

- ▶ Bradley & Terry. Rank Analysis of Incomplete Block Designs. *Biometrika*, 1952. Christiano et al. Deep RL from Human Preferences. NeurIPS 2017.
- ▶ Stiennon et al. Learning to Summarize from Human Feedback. 2020.
- ▶ Ouyang et al. Training Language Models to Follow Instructions with Human Feedback (InstructGPT). 2022.
- ▶ Bai et al. Training a Helpful and Harmless Assistant with RLHF. 2022.
- ▶ Gao, Schulman, Hilton. Scaling Laws for Reward Model Overoptimization. ICML 2023.

Modern alignment.

- ▶ Rafailov, Sharma, Mitchell, Ermon, Manning, Finn. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. NeurIPS 2023.
- ▶ Xu et al. Is DPO Superior to PPO for LLM Alignment? ICML 2024.
- ▶ Shao et al. DeepSeekMath (GRPO). 2024.
- ▶ Bai et al. Constitutional AI: Harmlessness from AI Feedback. 2022.

- ▶ DeepSeek-AI. DeepSeek-R1 / R1-Zero. *Nature* 645:633–638, 2025 (arXiv 2501.12948). Lambert et al. Tülu 3 (RLVR). 2024.

Multi-Agent RL.

- ▶ Lowe et al. MADDPG. 2017. Sunehag et al. VDN. 2018. Rashid et al. QMIX. 2018.
- ▶ Vinyals et al. AlphaStar (StarCraft II). 2019. OpenAI. Dota 2 with Large Scale Deep RL (OpenAI Five). 2019.

Robotics.

- ▶ Tobin et al. Domain Randomization. 2017. Akkaya et al. Solving Rubik's Cube with a Robot Hand (Dactyl). 2019.
- ▶ Hwangbo et al. Learning Agile and Dynamic Motor Skills for Legged Robots (ANYmal). 2019. Rudin et al. Massively Parallel Sim. 2022.
- ▶ Luo et al. SERL. 2024.

Systems.

- ▶ Mirhoseini et al. A Graph Placement Methodology for Fast Chip Design. *Nature*, 2021 (later named **AlphaChip**; Addendum, *Nature* 634:E10, 2024).

These slides have been adapted from Anna Goldie, *RL in the Real World: From Chip Design to LLMs*, guest lecture in **CS224R: Deep Reinforcement Learning**, Stanford.

- [1] Williams, R. J. (1992). Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8, 229–256.
- [2] Schulman, J., Levine, S., Moritz, P., Jordan, M. I., & Abbeel, P. (2015). Trust Region Policy Optimization. *ICML*.
- [3] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal Policy Optimization Algorithms. *arXiv:1707.06347*.
- [4] Shao, Z., Wang, P., Zhu, Q., et al. (2024). DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models (GRPO). *arXiv:2402.03300*.